

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY | GRADE 10 | 2.1 PLANT NUTRITION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the sub-strand. Think back to Amina's sukuma wiki in her Nairobi kitchen garden: it wilts at midday even when the soil is moist, recovers overnight, and turns pale and grows slowly when moved into the shade of a mango tree. Use evidence from all your lessons (leaf structure, what plants need for photosynthesis, the starch test, the light-dependent and light-independent reactions, limiting factors, and mineral nutrition) to answer each question in the space below it. Write in full sentences, use correct scientific terms, and link every answer back to the sukuma wiki. Where a question asks for a claim, give your evidence and reasoning. Where it asks for a diagram or table, draw it neatly.

Part 1: How do plants get their food?

Amina's sukuma wiki and a mushroom growing nearby get their food in different ways. (a) Name the type of nutrition each organism uses and explain the difference between them. (b) State what a green plant needs for photosynthesis: which of these are materials used up as reactants and which are not? For each, say how it reaches the leaf or is used there. (c) Write the word equation and the balanced chemical equation for photosynthesis.

Write your answer here:

Part 2: Leaf structure and the wilting puzzle

(a) Choose THREE leaf structures you saw in the cross-section of the mango or bean leaf. For each, describe how its structure suits its function. (b) Use what you know about stomata, transpiration, xylem and turgor to explain why Amina's sukuma wiki wilts at midday even though the soil is moist. (c) Explain why the plant recovers overnight.

Write your answer here:

Part 3: Evidence from the starch test and the two stages of photosynthesis

(a) In the starch test on a variegated or partly covered leaf, describe the iodine result in each of the three regions: white region, covered green region, and uncovered green region. What does each result tell you? (b) Describe what happens in the light-dependent reactions and in the light-independent reactions (Calvin cycle), including where each takes place and how the two stages are linked. (c) Use both stages to build a cause-and-effect chain that explains why the sukuma wiki in the shade of the mango tree grows slowly and turns pale.

Write your answer here:

Part 4: Limiting factors and Kenyan farming

(a) In the Lesson 7 investigation you counted oxygen bubbles at lamp distances of 5, 10, 20, 30 and 40 cm. Describe the shape of the graph of rate of photosynthesis against light intensity, and explain why the line flattens at high light intensity. What is a limiting factor? (b) Name the other factors you studied that affect the rate of photosynthesis, and describe how temperature changes the rate. (c) Greenhouse tomato farmers in Naivasha add CO₂, but adding CO₂ to the tea farms in Kericho would not help much. Explain why using the idea of limiting factors.

Write your answer here:

Part 5: Minerals and the complete explanation for Amina

(a) Name two mineral nutrients taken up from the soil, say how roots absorb them, and describe what each is needed for and what deficiency symptom you would expect. (b) Write a complete explanation answering the driving question: why does the sukuma wiki wilt at midday even when the soil is wet, and what does a plant actually need to stay green, grow, and make its own food? Your answer must explain both the wilting and the pallor in the shade. (c) Give one practical piece of advice, based on your explanation, that a Kenyan smallholder farmer could use to improve sukuma wiki yield.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of photosynthesis (reactants, light and chlorophyll, equations, two stages)	Correctly distinguishes the reactants from light energy and chlorophyll, gives both equations, and accurately describes where each stage occurs, the key steps and how ATP and NADPH link the stages.	States most requirements and equations correctly and describes both stages with minor omissions or errors.	Names some requirements or an equation but confuses the stages or leaves out key steps.
Explaining wilting using leaf structure and water movement	Clearly links stomata, transpiration, xylem supply rate and loss of turgor, explains why wet soil does not prevent wilting, and explains overnight recovery.	Explains wilting through water loss and turgor but is less clear about rate of loss versus supply or about recovery.	Says the plant lacks water or is hot without linking stomata, transpiration and turgor.
Use of evidence and limiting factors	Accurately uses the starch test, the light-intensity graph shape, the temperature curve and the definition of a limiting factor to justify claims, and correctly applies them to the Naivasha and Kericho examples.	Uses some evidence and defines a limiting factor, with partial application to the farm examples.	States facts without evidence, or the limiting factor idea is missing or incorrect.
Integration of minerals and the complete causal chain for the pallor	Builds a clear, sequenced chain from low light to less ATP and NADPH to less glucose to slow growth and less	Gives a mostly complete chain but one step or the link to chlorophyll or minerals is missing or unclear.	Gives a brief statement that shade means less food, without a causal chain.

	chlorophyll maintained, and explains the role of nitrogen and magnesium in chlorophyll.		
Communication and Kenyan application	Uses precise scientific vocabulary, organises answers clearly, and gives a sensible, evidence-based farming recommendation.	Mostly clear writing with some correct terms and a reasonable but general recommendation.	Writing is unclear or uses few scientific terms, and the recommendation is missing or not linked to the evidence.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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